Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation

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Abstract

Automated research and self-improvement pipelines choose a reward model or LLM judge by its accuracy on a validation set. However, the pipeline or its designers often decide which items that set contains, so whoever curates the set can also choose the evaluator. We introduce *curation certificates*, which state how much of the validation set could be removed before a different model wins. For benchmarks scored by weighted accuracy, the certificate in benchmark weight is at least the score gap between the top two models and, for binary scores, within one item's weight of it; the certificate in items is computed exactly by sorting per-item score differences. RewardBench's winner (118 models, 2,985 items) stays first when any 11 items are removed, but removing 12 particular items puts the runner-up first, and at a 1% curation budget any of 3 models could be the winner. RewardBench 2's winner (188 models, 1,753 items) stays first when any 42 items are removed. On random 200-item RewardBench validation sets, the 8% of selections whose top-two gap exceeds 2 points match the full-benchmark winner 87% of the time, against 43% for the rest. We give a protocol for reporting each selection with its certificate.

1 Introduction

Systems that automate research or improve themselves need an evaluator: a reward model to optimize against, or an LLM judge that ranks candidate outputs (Lu et al., 2024; Zheng et al., 2023). Evaluators are chosen by meta-evaluation: each candidate is scored by how often it agrees with human preference labels on a benchmark such as RewardBench (Lambert et al., 2025; Malik et al., 2026), and the top scorer is selected. However, in an automated pipeline the pipeline itself often picks the items used for this choice, as a filtered subset, a domain slice or a small validation set. If the ranking of evaluators depends on which items are kept, then whoever picks the items also picks the evaluator, which is reward hacking one level above the reward (Gao et al., 2023). This raises the question we study: **how much of the validation set must be removed before a different evaluator wins?**

Leaderboard rankings are known to shift with the choice of benchmark, the answer format and private testing (Dehghani et al., 2021; Alzahrani et al., 2024; Singh et al., 2025; Zhang and Hardt, 2024), and with targeted edits to individual items (Dubrovsky et al., 2025). Gordienko et al. (2026) count how many *datasets* a developer must train on to top a multi-task leaderboard, and show that this problem is NP-hard under mean win rate. Oyarhoseini et al. (2026) use influence approximations to show that editing under 1% of pairwise matches can change the leader of a Bradley–Terry arena. Both papers study how a ranking can be attacked. For training data, related certificates exist: dropping a small fraction of observations can overturn a statistical conclusion (Broderick et al., 2020), and certified defenses bound the damage that poisoned training points can do (Steinhardt et al., 2017). We give the analogous certificate for the evaluation data used to select a model, one that whoever selects an evaluator can compute and report, for the weighted-accuracy scores used to meta-evaluate reward models and judges. We contribute two curation certificates for the winner of such a benchmark, a lower bound in benchmark weight and an exact count in items, both computed from the per-item score table alone (§3); a proof that the weight bound is tight to within one item's weight for binary scores, with evidence from linear programs beyond the runner-up; and two uses, possible-winner sets under a curation budget and a check on selections from small validation sets (§4).

2 Scores, Curations and Data

A benchmark has items $i = 1, \dots, N$ with official weights $w_i^0 \geq 0$, $\sum_i w_i^0 = 1$. Model m has per-item score $r_{mi} \in [0, 1]$ and official score $S_m = \sum_i w_i^0 r_{mi}$. A *curation* keeps a subset K of the items, with kept share $f = \sum_{i \in K} w_i^0$, and rescores every model on K alone, with the kept items' relative weights unchanged: $S_m^K = \sum_{i \in K} w_i^0 r_{mi} / f$. We call $1 - f$ the *removed share*, and report scores, gaps and shares in percentage points. K *crowns* m if m scores strictly highest on it, $S_m^K > S_k^K$ for every $k \neq m$. The winner is *certified* against a class of curations if it stays first under every curation in the class.

Data. RewardBench v1 publishes per-item results for 118 models on 2,985 preference pairs in 23 subsets (Lambert et al., 2025). We rebuild each official score from them: a section score is the count-weighted mean of its subsets' accuracies, with `math-prm` upweighted as in the released code, and the overall score is the mean of the four section scores. For four models, from the winner down to a 0.5B chat model, our subset accuracies match RewardBench's published values exactly. The result files list items by position, and a check of the item text found no misaligned items. RewardBench 2 (Malik et al., 2026) publishes results by item id. We use the 188 models with results on all 1,753 items of its five accuracy subsets, score each model by the mean of its subset accuracies, and exclude the Ties subset, which uses a different metric, so our scores differ slightly from the official ones.

3 Certificates in Weight and in Items

Let a be the winner. For each challenger c, let $g_c = S_a - S_c > 0$ be its gap to the winner and $d_i^c = w_i^0 (r_{ai} - r_{ci})$ the contribution of item i to that gap.

Proposition 1. *Any curation K on which c ties or beats a ($S_c^K \geq S_a^K$) removes a share of at least g_c. Hence the winner is certified against every curation that removes a share below $g^\star = \min_c g_c$, its gap to the runner-up.*

Proof. $f\,(S_a^K - S_c^K) = \sum_{i \in K} d_i^c = g_c - \sum_{i \notin K} d_i^c \geq g_c - (1 - f)$, since $d_i^c \leq w_i^0$; this is positive whenever $1 - f < g_c$. □

Proposition 2. *Sort the contributions so that $d_{(1)}^c \geq d_{(2)}^c \geq \cdots$, and let $k_c = \min\{k : \sum_{j \leq k} d_{(j)}^c > g_c\}$, with $k_c = \infty$ if no such k exists. Removing a set R of items makes c overtake a if and only if $\sum_{i \in R} d_i^c > g_c$. So no removal of fewer than k_c items lets c overtake a, and removing the k_c items with the largest contributions does. With $k^\star = \min_c k_c$, no challenger overtakes the winner when any $k^\star - 1$ items are removed, and some $k^\star$ items let one do so, so this count is exact.*

Proof. From the identity above, $S_c^K > S_a^K$ iff $\sum_{i \notin K} d_i^c > g_c$. Any k removed items contribute at most $\sum_{j \leq k} d_{(j)}^c$. □

Both certificates need only the table of per-item scores and no extra model calls: $g^\star$ can be read off the leaderboard, and $k^\star$ takes one sort per challenger. For binary scores, the weight certificate is also nearly exact:

Proposition 3. *Let a and the runner-up c have scores in $\{0, 1\}$, let every other model score strictly below c, let c beat a on some item, and let $w_{\max} = \max_i w_i^0$. Removing items with $r_{ai} = 1$, $r_{ci} = 0$ one at a time until their weight exceeds $g^\star$ makes c the strict winner, at a removed share of at most $g^\star + w_{\max}$. Removing the k_c items of Prop. 2 also does.*

Proof. The weight of those items minus the weight of the items c wins is $g^\star$, so they weigh more than $g^\star$, and each removal adds at most $w_{\max}$. Then c overtakes a by Prop. 2. As $r_{ci} = 0$ on every removed item, $f S_c^K = S_c$, while $f S_m^K \leq S_m < S_c$ for every other m. The k_c items all have $d_i^c > 0$, so they are such items. □

Cheapest crowning curations. Beyond the runner-up, Prop. 1 is only a lower bound, because removing items that favor the winner over one challenger can let a third model overtake both. To see how close the bound is, we compute for each model m the largest kept share f at which some curation crowns it. For fixed f, relaxing the subset to item weights w gives a linear program:

$$\max_{w,t} \; t \;\; \text{s.t.} \;\; \sum_i w_i (r_{mi} - r_{ki}) \geq t \;\; \forall k \neq m, \qquad 0 \leq w_i \leq w_i^0 / f, \quad \sum_i w_i = 1,$$

where every subset with kept share f is a feasible w, so $t^* \leq 0$ proves that no such subset crowns m. If $t^* > 0$, we round the solution to an item subset, keeping the items whose LP weight exceeds half their upper bound or, if that subset fails, every item with positive weight, and accept it only

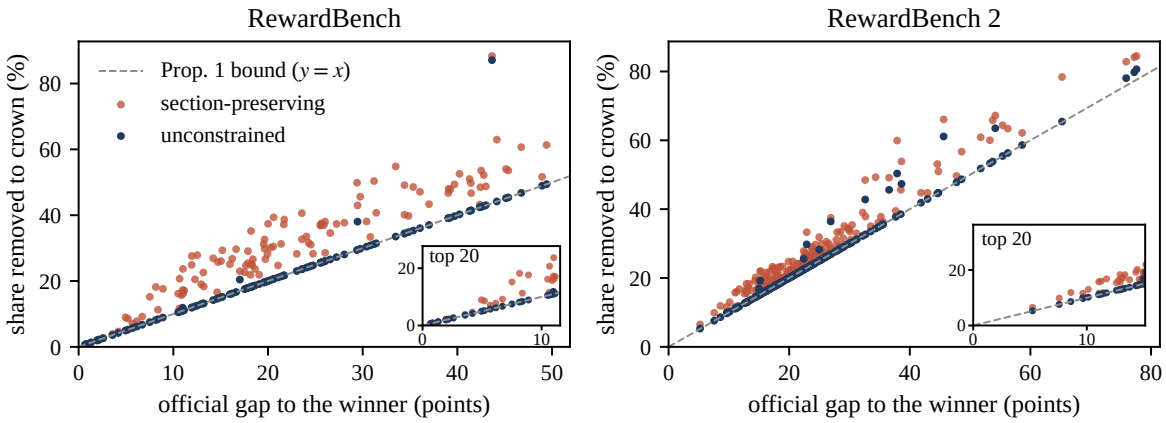


Figure 1: Share of the benchmark that must be removed to crown each model, against the model's official gap to the winner: RewardBench (118 models, left) and RewardBench 2 (188, right). Blue: unconstrained curation. Orange: curation that holds each section's share fixed, to which Prop. 1 does not apply (see Limitations). Insets: the top 20 models. Blue points lie on the bound $y = x$ of Prop. 1 for most models, so the top-two gap is close to the true cost of changing the winner.

if m is strictly first after rescoring. We scan f downward from 1 and then bisect. Every reported cost is the removed weight of such a verified subset, and so an upper bound on the cheapest crowning removal, which Prop. 1 bounds from below. A *section-preserving* variant also fixes each section's total weight, rescaling the kept items within each section, so that the curated set keeps the benchmark's composition.

Possible-winner sets. Let the curation budget δ be the largest share that may have been removed, and let $P(\delta)$ be the set of models that some curation removing at most δ can crown. By Prop. 1, $P(\delta)$ contains only models within δ of the winner, $P(\delta) \subseteq \{m : S_a - S_m \le \delta\}$. Every model whose verified LP cost is at most δ is in $P(\delta)$. Together these bound the size of $P(\delta)$ from both sides.

4 Results on RewardBench and RewardBench 2

RewardBench's winner falls to 12 removed items. RewardBench's winner leads by $g^\star = 0.67$ points, so it stays first under any curation that removes less than 0.67% of benchmark weight. Its heaviest item weighs 0.07 points, so by Prop. 3 a removal of at most 0.74% dethrones it, and the cheapest curation we found removes 0.68%. In items, the winner stays first when any 11 items are removed, and by Prop. 3 removing 12 particular items makes the runner-up the new winner. RewardBench 2's winner leads by 5.22 points, loses first place after 5.30% of the weight is removed, and stays first when any 42 items are removed.

Budget δ	0.5%	1%	2%	5%	10%
RewardBench	1	3	5–6	11	19
RewardBench 2	1	1	1	1	6

Table 1: Possible winners $|P(\delta)|$ at curation budget δ (% of benchmark weight). In a range a–b, a models have a verified curation that removes at most δ and b models are within δ of the winner (Prop. 1). A single number means $a = b$.

The weight bound is nearly tight. Every model on both benchmarks can be crowned by some curation. On RewardBench, the ratio of removed share to gap has a median of 1.00 and is at most 1.08 among the top 20 models (Figure 1); the largest ratio over all models is 2.00. On RewardBench 2, the median ratio is 1.00 and the maximum 1.35. The pool includes LLM judges: GPT-4o (rank 18) can be crowned by removing 8.41%, so it is a possible winner at a 10% budget. For a curator who also keeps each section's share fixed, so that the benchmark's composition is unchanged, the cheapest curations we found cost a median of 1.39 times the gap on RewardBench and 1.12 on RewardBench 2; these are upper bounds. Because the gap bounds the cheapest unconstrained removal from below and each reported cost bounds it from above, a ratio near 1 pins that removal down from both sides.

	RewardBench			RewardBench 2		
n	all	cert.	not	all	cert.	not
100	35	64 (15)	30	87	95 (76)	61
200	46	87 (8)	43	97	100 (88)	75
500	66	98 (3)	65	100	100 (97)	97

Table 2: Selection on random validation sets of n items, drawn with replacement in proportion to official weight (2,000 draws per n). Each entry is the % of draws that select the full-benchmark winner: over all draws (all), over draws whose top-two gap on the set exceeds 2 points (cert.; % of draws in brackets), and over the remaining draws (not).

At a 1% budget, 3 RewardBench models could win. Because the bound is nearly tight, the two ends of $P(\delta)$ are close (Table 1). On RewardBench, a pipeline that may have curated 1% of the benchmark can only say that the best model is one of 3, and at 5% one of 11. RewardBench 2's winner is the only possible winner up to a 5% budget.

The top-two gap flags unreliable validation-set selections. In practice a pipeline scores candidates on a small validation set, and small item sets can estimate benchmark scores well (Maia Polo et al., 2024); the certificate is then the top-two gap on that set. Table 2 draws such sets at random, so that each set's score is an unbiased estimate of the official score. These draws measure sampling noise, not curation. Props. 1–2 applied to that set cover curation of the set itself, so the top-two gap is a guarantee against curation and, empirically, a signal of sampling noise. On RewardBench with 200 items, the 8% of selections certified at 2 points (151 of 2,000) match the full-benchmark winner 87% of the time (Wilson 95% interval 81–92%), against 43% for the rest (41–45%). At 1 point the split is 66% against 35%, and at 5 points no selection is certified, as expected when the full-benchmark winner leads by 0.67 points. On RewardBench 2, whose winner leads by 5.22 points, most selections are certified and nearly all are correct; at 5 points the split is 100% against 94%.

5 Protocol for Certified Selection

The certificates cover removals from a committed item set, not choices made before it was fixed, and they need per-item scores on all of it. For an automated pipeline that selects an evaluator: (i) fix the validation items and record a hash of them before scoring any candidate, score every candidate on all of them, and log every filter applied afterwards; (ii) set a curation budget δ, the share of the set that could have been dropped or filtered; (iii) report the certificates $g^\star$ and $k^\star$ with the selection, and accept the winner only if $g^\star > \delta$; (iv) otherwise, report the set of possible winners, bounded as above, instead of a single winner. On RewardBench with a 1% budget, step (iv) applies, and the pipeline reports 3 possible winners.

6 Conclusion

We introduced curation certificates for selecting a reward model or LLM judge by weighted accuracy: a lower bound in benchmark weight, equal to the top-two gap and tight to within one item's weight for binary scores, and an exact count in items. On small validation sets, the same gap separates reliable selections from unreliable ones. We hope that automated pipelines will report a certificate with every evaluator they select.

Limitations

Removal only. A curator who could also reweight the kept items, paraphrase them or change the prompt format would have more freedom, so the certificates hold only against removal.

Fixed relative weights. Props. 1–3 assume that kept items keep their relative weights. A scorer that recomputes section means after removal raises the kept items' weights, so the weights change by twice the removed share in total and the floor falls to $g^\star/2$. Such a pipeline should require $g^\star > 2\delta$, and $k^\star$ does not apply.

Scope. The certificates apply to weighted accuracy, not to ordinal aggregates such as mean win rate. We do not claim that any published result was curated.

Reproducibility and AI Use

All results come from public per-item result files (`allenai/reward-bench-results`, `allenai/reward-bench-2-results`). All analysis ran on one laptop CPU with SciPy's HiGHS solver, and no models were run. Code and derived data are at `https://github.com/adrian-erlikhman/curated-to-win`. An AI assistant (Claude, Anthropic) helped write the analysis code and draft the text. The author chose the study, checked the proofs and the results, and takes responsibility for the content.

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